

Admin dashboard: workflow audit

Huchu schools vertical audit · Admin dashboard: workflow audit · admin/workflows.md · rendered 2026-09-14

Surface: configuration and master data for a school tenant. Pages `app/management/master-data/schools/{identity, years, periods, classes, subjects, grading}`, the `app/schools/academics/*` redirects, `app/schools/{calendar, timetable, teachers, teachers/assignments, staff, imports, documents, transport, library, boarding/hostels, notices}`, guardian and student record pages for portal invites and custom fields, and the tenant-level provisioning that seeds a school (`lib/schools/provision.ts`). APIs under `app/api/v2/schools/{academic-years, terms, periods, rooms, classes, streams, subjects, syllabus, grading-schemes, calendar, settings/identity, field-definitions, staff, teachers/**, timetable/**, guardians, guardian-links, portal-invites, imports/**, library/**, transport/**, teaching-resources, lesson-plans, notices, messages}`. Personas: tenant `SUPERADMIN` and `MANAGER` (unconstrained), `SCHOOL_ADMIN` (full school grants), `REGISTRAR` (academics create and edit, students, teachers).

Audit date: 2026-09-14. Static read of `main`, cross-checked against the roadmap (Iterations 0, 1, 3, 4; S-1.1, S-1.2, S-1.7, S-1.16, S-1.17, S-3.1, S-3.3, S-4.4, S-4.5, S-5.x), the production-readiness audit, the schools implementation plan, and the K-12 benchmark §2, §3, §4, §10, §12, §13, §16, §20. UI companion: `ui-ux.md` in this folder.

1. Verdict in one paragraph

Setup is the most complete layer of the schools vertical and the least visible. A school can be provisioned with a year, terms, a class ladder, subjects, a grading scheme, a fee structure and roles in one operation; the office can then set periods and rooms, build a clash-checked timetable by hand or with the greedy filler, keep a calendar that drives "not a school day", link teachers to HR, add custom fields, search every record, import a previous system's data with a dry run and a rollback, and print eight document types. What lets it down is reachability and finish. The master-data ladder lives under Management with its own rail and is missing from the school sidebar; calendar events cannot be edited; four of the eight documents have no page; teacher and guardian accounts have no self-service path (invites are copied by hand, teachers get no invite at all); notices reach only families with a claimed portal account; and the platform pieces a school administrator expects, tenant branding, communication channels, a setup checklist, multi-campus, are not there.

Runtime check (14 September 2026)

Verified on the seeded St Marys tenant (see `../reference/runtime-verification.md`). Confirmed at runtime: provisioning and the demo seed build a working school (year, terms, ladder, subjects, structures, invoices); the sidebar drops the master-data pages (B1); the opening-balance import guard refuses a bursar (B8); the guardian-link PATCH has no persona check (B5). New observation: the portal host prefixes did not serve the portal login on this setup (see open decisions).

2. Docs versus code

Claim	Source	Code reality	Verdict
S-0.1 academic year and terms, one current: done	roadmap	Partial unique indexes and transactional activation (<code>lib/schools/calendar.ts:179-204</code>).	True.
S-1.1 timetable with clash rejection and copy-forward; S-1.16 auto-fill; S-1.17 bulk allocation: done	roadmap	All present (<code>lib/schools/timetable.ts</code>). The timetable page cannot place a lesson by clicking the grid (UI audit).	True.
S-1.2 calendar and holidays: done	roadmap	Create and delete only; no PATCH /calendar/[id] ; editing is delete and recreate (<code>school-days-content.tsx:133, 169, 192</code>).	Partial.
S-1.7 teacher linked to Employee deliberately: done	roadmap	<code>teacher-identity.ts</code> suggests and links on click; "Find the employee" on every row.	True.
S-3.1 provisioning seeds the ladder: done	roadmap	<code>provisionSchool</code> plus <code>ensureCurrentTermEnrolments</code> . Open-questions #8 and #17 still say provisioning writes no enrolments.	True; open-questions stale.
S-3.3 import with dry run, idempotent re-run, rollback: done	roadmap; \$199 add-on	Present with audit rows. Opening balances require <code>students:create</code> and <code>fees:create</code> together, so only SCHOOL_ADMIN can import them; cross-currency balances import at rate 1 and are flagged; no partial-commit undo.	True with gaps.
S-4.4 custom fields on school record types: done	roadmap	Present for student and guardian; <code>configure</code> is SCHOOL_ADMIN only; no fields tab on teacher.	True.
S-4.5 unified search: done	roadmap	⌘K over students, guardians, staff, classes, subjects, hostels.	True.
S-5.1 to S-5.3 eight document sources: done	roadmap	Sources exist; <code>/schools/documents</code> surfaces report card, invoice, class list and register only (<code>school-documents-content.tsx:770-773</code>). Receipt, statement, admission letter and transfer letter print from record rows if at all.	Engine done, surface half done.

Claim	Source	Code reality	Verdict
S-0.3 invites from detail pages or in bulk: <code>done</code>	roadmap	Present. Links shown once and copied by hand; guardians without an email are skipped (<code>portal-invite-dialog.tsx:64-66</code>); no email, SMS or WhatsApp send; no teacher invite.	True and operationally weak.
S-9.6 notice entity with audience targeting: <code>todo</code>	roadmap	<code>POST /notices</code> with audience, class or pupil shortlist and severity exists; in-app only; no draft, schedule or recall.	Built more than the row says.
Marketing: "Custom branding and school domain" (Premier and a \$79 add-on)	pricing	<code>lib/platform/tenant.ts</code> carries no theme fields; portals show <code>companyLabelFromHost</code> only; no logo or colour anywhere. Domain provisioning exists at platform level.	Domain yes, branding no; the band and the add-on contradict each other.
Marketing: "Unlimited staff and teacher accounts"	pricing	No user-count enforcement found in the school code; the platform has a <code>USER_PACK_SIZE</code> concept.	Unverified; check entitlement.
Marketing: multi-campus consolidation (Group band)	pricing, site FAQ	One <code>companyId</code> is one school; no campus entity.	Overclaim.
Production readiness: "34 pages"	Aug 4 audit	69 page files (14 redirects).	Stale.
<code>live-capabilities.md</code> lists <code>/schools/portal/*</code> , <code>/schools/assessments</code> as live pages	system reference	Redirects; assessments folded into results.	Stale.
Pack spec roles <code>academic-admin</code> , <code>timetable-admin</code> , <code>communications</code>	pack spec	Not in the role model.	Stale.

3. Workflow inventory

Tenant and provisioning

#	Workflow	Status	Evidence and gap
A1	Provision a school tenant with year, terms, ladder, subjects, grading, fee structure, roles	Implemented	<code>lib/schools/provision.ts</code> ; operator wizard on <code>TEMPLATE_SCHOOLS</code> . Schools remain operator-provisioned by policy.
A2	Feature entitlement for the pack and portals	Implemented	<code>ADDON_SCHOOLS_SUITE</code> bundle; band-to-feature mapping (Community without portals, Standard without teacher portal) is not expressed anywhere.

#	Workflow	Status	Evidence and gap
A3	School identity: student-number format, ID-card design, presentation	Implemented	Only page that gates its own controls by role. No card print or issue run.
A4	Branding: crest, colours, portal theming	Missing	Sold.
A5	Multi-campus	Missing	Sold in the Group band.
A6	Setup checklist and readiness	Missing	A new tenant lands on an overview of zeros.

Academic structure

#	Workflow	Status	Evidence and gap
A7	Academic years and terms, make current	Implemented	Terms created one at a time; "Make current" has no confirm although it flips every screen.
A8	Calendar: holidays, events, teaching-day override, Zimbabwe public holidays	Partial	No edit route; weekends closed by default with no per-school teaching-days setting (open-questions #2).
A9	Classes, streams, capacity, form teacher	Implemented	Streams are a second tab, not rows under their class.
A10	Subjects, syllabus (scheme of work), core or elective, pass mark	Implemented	Subjects can be created from two places (master data and the teachers page).
A11	Grading schemes and bands, default scheme	Implemented	Overlapping bands refused in code, not DB (open-questions #3).
A12	Publish windows	Implemented	Configured in two places.
A13	Periods and rooms	Implemented	No "generate eight periods from 07:30 every 40 minutes".
A14	Custom fields on student and guardian	Implemented	None on teacher; ownership of "teaching qualification" undecided.
A15	Houses, departments as entities, option blocks	Missing	Department is a string on the teacher profile.

Timetabling

#	Workflow	Status	Evidence and gap
A16	Manual slot placement with class, teacher and room clash rejection	Implemented	UI is a sheet with six pickers; no click-a-cell.
A17	Auto-fill (greedy first fit) reporting what it could not place	Implemented	
A18	Copy forward from last term	Implemented	
A19	Cover arrangement	Implemented	Office-only by design; a teacher cannot request it.

#	Workflow	Status	Evidence and gap
A20	Exam timetable and invigilation, room booking	Missing	
A21	Publish the timetable to portals with change notifications	Partial	Portals read it; no notification on change.

People

#	Workflow	Status	Evidence and gap
A22	Teacher profiles, subjects, assignments (single and bulk), HOD and class-teacher flags	Implemented	Profile DELETE is a hard delete (<code>teachers/profiles/[id]/route.ts:201</code>).
A23	Link a teacher to an HR employee	Implemented	Per row; no bulk link.
A24	Support staff via HR with a school role label	Implemented	<code>staff/route.ts</code> re-doors <code>/api/employees</code> .
A25	Teacher portal account	Partial	Office creates a <code>User</code> and a profile with <code>userId</code> ; no invite, no password reset for anyone but SUPERADMIN.
A26	Guardians: create, link to children, consent flags, primary	Implemented	<code>PATCH /guardian-links/[id]</code> has no persona check; guardian DELETE is a hard delete.
A27	Portal invites for guardians and students, revoke, re-issue	Implemented	Manual link copy; no channel; no email means no invite; claim can reset an existing account (see portal audits).
A28	Staff leave, appraisal, CPD for teachers without an HR record	Missing	HR module covers employees only.

Data and documents

#	Workflow	Status	Evidence and gap
A29	Import classes, students, guardians, fee structures, opening balances with dry run, commit, rollback	Implemented	Role composition gap for fee entities; 5,000-row cap; no partial undo.
A30	Export	Partial	Reports CSV and PDF; no full data export per entity.
A31	Documents: report card, invoice, class list, register from a page; receipt, statement, admission letter, transfer letter from rows	Partial	Four sources without a page; no template editing in the school shell; no batch by class for statements.
A32	ID cards	Missing	Design settings only.
A33	Data quality: pupils not in a class, guardians without a phone, duplicates	Missing	Only the admissions duplicate check.

#	Workflow	Status	Evidence and gap
A34	Audit log of master-data changes	Missing	Academic-structure routes write no audit rows.

Services and communication

#	Workflow	Status	Evidence and gap
A35	Hostels, rooms, beds	Implemented	Beds added per room dialog; no "add 24 beds".
A36	Library catalogue, copies, loans, fines	Implemented	Fines never billed; no ISBN lookup or barcode entry.
A37	Transport routes, stops, riders, boarding register	Implemented	Billing reported, never posted.
A38	Notices with audience and severity	Implemented	In-app only; reports:create so SCHOOL_ADMIN only; no draft, schedule or recall.
A39	Communication channels (email, SMS, WhatsApp) configuration	Missing	lib/notifications.ts has no school emitters.
A40	Integrations: ZIMRA fiscalisation, GL posting	Implemented	The only two. No payment gateway, no LMS, no exam board.

Counts: 24 implemented, 7 partial, 0 broken, 9 missing.

4. Benchmark gap

Capability	Benchmark norm	Huchu today	Priority
Academic structure, calendar, grading, timetable with constraints and auto-fill	Standard	Present, strong	Done
Data import with dry run and rollback	Standard	Present	Done
Custom fields and unified search	Standard	Present	Done
Setup checklist and data-quality report	Standard	Absent	P1
Account provisioning with delivered invites and self-service reset	Standard	Manual links, no reset	P0
Communication channel configuration (SMS, WhatsApp, email)	Standard	Absent	P0
Tenant branding on portals and documents	Sold	Absent	P1
Exam timetable, room booking	Standard	Absent	P2
Departments and houses as entities	Standard	Absent	P2

Capability	Benchmark norm	Huchu today	Priority
Audit log of configuration changes	Standard	Absent	P1
ID card issuing	Common	Design only	P2
Multi-campus	Sold	Absent	P2 (decision)
Staff leave and cover requests for teachers	Standard	Absent for non-HR teachers	P2

5. Bugs and risks

#	Finding	Location	Severity
B1	Master-data pages are not in the school sidebar; the only "Academic setup" entry that survives the workspace mapping is "Scheme of work", which redirects to the teacher portal.	<code>lib/workspaces.ts:410-419; app/schools/academics/syllabus/page.tsx:11</code>	High
B2	No password reset for portal or staff users except a SUPERADMIN route; no teacher invite.	<code>app/api/users/password-reset/route.ts:26; portal-invites.ts:82</code>	High
B3	Invite delivery is copy-paste; guardians without email cannot be invited.	<code>portal-invite-dialog.tsx:64-66, 117-133</code>	High
B4	Claim can reset the password and demote the role of an existing same-tenant account.	<code>lib/schools/portal-invites.ts:243-273</code>	Medium
B5	<code>PATCH /guardian-links/[id]</code> lacks a persona check.	<code>guardian-links/[id]/route.ts:46-110</code>	Medium
B6	Hard deletes for students, guardians and teacher profiles behind "archive".	<code>students/[id]/route.ts:379; guardians/[id]/route.ts:253; teachers/profiles/[id]/route.ts:201</code>	Medium
B7	Calendar events cannot be edited.	<code>app/api/v2/schools/calendar/[id]/route.ts</code> (DELETE only)	Medium
B8	Import guard requires <code>students:create</code> and <code>fees:create</code> together.	<code>imports/_guard.ts:19-24</code>	Medium
B9	Grading-schemes API is behind <code>schools.results</code> while its page is behind <code>scho ols.core</code> ; a core-only tenant sees the page and gets 403s.	<code>route-registry.ts</code>	Low
B10	No audit rows for academic-structure changes (years, terms, classes, subjects, grading, timetable, calendar).	all setup routes	Medium

#	Finding	Location	Severity
B11	Notices reach only families with claimed accounts; withoutAccount count is reported and dropped.	lib/schools/notices.ts:169-199	Medium
B12	Route-guard coverage test checks file-level markers only, so a method without a check passes (B5).	lib/schools/route-guard-coverage.test.ts:113	Low
B13	Open-questions #6, #7, #8, #17 describe states the roadmap closed; production readiness §6 lists built modules as absent.	docs	Doc hygiene

6. Proposed edits

1. **Sidebar (B1).** Generate the school sidebar from lib/navigation.ts so the Setup band appears (Years and terms, Calendar, Classes and streams, Subjects, School day and rooms, Grading and windows, Records and identity); remove the syllabus redirect from the nav.
2. **Accounts (B2, B3, B4).** Add a forgot-password flow for portal and staff hosts (token by email or SMS); add a teacher invite subject; send invite links through a channel; allow phone-number invites with an SMS or WhatsApp claim link; refuse claims that would overwrite a non-portal account.
3. **Persona check on guardian-link PATCH (B5);** make the route-guard test per method (B12).
4. **Soft archive (B6)** and PATCH /calendar/[id] (B7).
5. **Import guard per entity (B8);** align the grading-schemes feature key with its page (B9).
6. **Audit setup changes (B10)** with PlatformAuditEvent on year, term, class, subject, scheme, window, timetable and calendar writes; a "who changed what" view under Setup.
7. **Notices (B11).** Show the withoutAccount families with a "send by SMS" option once a channel exists; until then, an export of who was not reached.
8. **Docs (B13).** Close the stale open-questions entries; correct production-readiness §6; update roadmap rows S-1.2 (no edit), S-5.1 (four sources without a page), S-9.6 (built in-app).

7. Proposed restructuring

- **A Setup band inside the school shell** (/schools/setup/*) replacing the Management → Master Data detour, ordered as imports need it: year → terms → classes → streams → subjects → school day and rooms → grading → publish windows → fee structures → staff. One rail, one shell.
- **One home for each master-data entity.** Subjects only under Setup; publish windows only under Grading; assignments only under People, with the teachers page linking there.
- **People band:** Teachers, Support staff, Assignments, Accounts (invites, portal status, resets) so account administration is a screen, not a per-row dialog.
- **Communication band** with a Channels settings page (sender ids, templates, costs) once a channel exists.
- **Persona-aware rail:** the admin persona sees Setup first; registrar sees Roll first.

- **Retire the pack spec's role list and feature keys** and the `verticalDataViews` phase specs; the roadmap and this audit are the record.

8. Proposed new workflows and features

1. **Setup checklist and readiness dashboard** (`/schools/setup`): steps done and outstanding with links; data-quality queue (pupils without a class, guardians without a phone, teachers without a portal account or HR record, classes without a form teacher, subjects without a teacher); last import with rejected rows and Undo.
2. **Account administration**: invite by email or phone with delivered links, resend, reset password, disable, "who has never signed in"; teacher invites; bulk invite by class with a delivery report.
3. **Communication channels**: SMS aggregator and WhatsApp Business sender configuration per tenant, templates with merge fields, cost tracking, delivery status; the `schools` notification category (S-9.5) with quiet hours.
4. **Tenant branding**: crest, colours, portal accent and document header stored on the tenant; applied to portals, login pages and PDFs. This is what the Premier band and the add-on sell.
5. **Calendar editing and a term-dates template**: edit events; create three terms with half-terms and public holidays from a template.
6. **Timetable builder on the grid**: click a cell to place, drag to move, clash highlighting; exam timetable and room booking as a second mode.
7. **Departments and houses** as entities: department on subject and teacher, HOD per department (feeds the moderation fix), houses for pastoral and sport.
8. **Document centre**: all eight sources with batch runs by class (statements, class lists, registers, ID cards), template editing per school, a print queue.
9. **Configuration audit log** with diff and actor.
10. **Multi-campus** as `SchoolCampus` under one company, if the Group band stays: per-campus registers, structures and staff with group reports.
11. **Staff self-service for teachers without an HR record**: leave request, cover request, profile edits, feeding the head's "cover today".

9. Open decisions

- Do schools stay operator-provisioned, or does the setup checklist become a self-serve onboarding?
- Which communication channel is first: SMS aggregator, WhatsApp Business, or email? The invite flow and notices both wait on it.
- Is branding an add-on or included? The pricing file says both.
- Multi-campus: build it or stop selling it.
- Who owns "teaching qualification" and other teacher attributes: school custom fields or HR?
- Portal host routing: on the local runtime pass the `staff.`, `parents.` and `students.` hosts served the tenant workspace sign-in at `/login` rather than the portal form, and the repository's own e2e fixtures record the same under host nomination and drive portals by internal path. Confirm the prefix rewrite works in production and add a test for it.